PROJECT PORTFOLIO | DATABASE DESIGN & REPORTING

Terranova 100-Site 4G Rural Rollout

Relational Database & Automated Reporting System

Microsoft Access · SQL · Visual Basic (VBA) · Data Modelling · Report Generation

1. Project Title & Overview

Terranova is a normalized relational database and reporting system built to plan, cost, and track a 100-site rural 4G network rollout across four provinces of northern Spain (A Coruña, Asturias, Lugo, and Pontevedra), serving an estimated 99,320 inhabitants.

The project replaces scattered spreadsheets with a single source of truth that connects site data, regulatory and permitting status, infrastructure, power, transmission and RF design, a 45-item unit-cost catalogue, and an 11-milestone schedule for every site. On top of this model sit parameterized SQL queries, a multi-subreport Site Profile report, and a Visual Basic automation layer that turns raw project data into management-ready dashboards and recurring reports.

Project at a glance

Metric	Value
Sites modelled	100 (4 provinces, 98 municipalities)
Population served	99,320 inhabitants
Database tables	19 (9 core + 10 lookup); 16 defined relationships
Cost lines (site × catalogue item)	4,600 (46 per site)
Unit-cost catalogue	45 items across 8 cost areas, in EUR and USD
Schedule records	1,100 (11 milestones × 100 sites, planned vs. actual)
Total modelled CAPEX	€9.90 M / US$11.48 M (catalogue-priced)
Saved analytical queries	7, including a parameterized search
Report	Site Profile report with 7 subreports

2. Role & Responsibilities

Role: Database Designer & Project Reporting Lead

- **Design.** Conceived the data architecture: entity-relationship model, normalization to third normal form, naming conventions, lookup tables, primary and foreign keys, and the cost-model redesign described below.
- **Planning.** Defined the reporting requirements and KPIs (CAPEX by area, province, and site; schedule variance by milestone; permit and infrastructure readiness) and mapped each to a query, report, or dashboard view before building.
- **Execution.** Built the tables, relationships, SQL queries, report layouts, and VBA automation; loaded and reconciled the data; validated totals.
- **Regular reporting.** Produced recurring status reports and refreshed dashboards as new data arrived, translating findings into conclusions and recommended actions for project stakeholders.

3. Challenge / Problem

A rural rollout of 100 sites is a data-integration problem as much as an engineering one. Each site carries information owned by different teams: land and negotiation, permits, civil works, power, transmission, RF, and regulatory fees.

1. **Fragmented data.** Cost, schedule, and technical attributes lived in separate sheets, making site-level and portfolio-level answers slow and error-prone.
2. **Unstructured cost data.** The first design stored roughly 27 cost values as separate columns across five domain tables, with the unit-cost catalogue unlinked to them. Totals could not be traced back to a price and a quantity.
3. **No consistent view of schedule performance.** Planned and actual dates for 11 milestones per site had to be compared systematically to expose delays early.
4. **Repeatable reporting.** Stakeholders needed the same trusted numbers every reporting cycle, without rebuilding reports by hand.

Goal: a normalized, auditable database with query-driven dashboards and automated reports that answers where the money is going, what is late, and which sites need attention in seconds.

4. Actions / Process

4.1 Methodology

- **Requirements-first design** aligned with PMBOK knowledge areas (cost, schedule, scope), so every table and query traces back to a project-management question.
- **Relational modelling** with an ERD, normalization to 3NF, surrogate keys, and lookup tables that enforce consistent vocabularies (owner type, access type, structure type, availability status, transmission medium).
- **Structured improvement cycle** (define, measure, analyze, improve, control), applied to the reporting process itself.

4.2 Steps performed

1. **Data modelling.** Designed 9 core tables (tblSites, tblDates, tblRegulatory, tblPermits, tblInfrastructure, tblElectrical, tblNetwork, tblUnitCosts, tblSiteCostLine) and 10 lookup tables linked by foreign keys.
2. **Cost-model redesign.** Replaced about 27 scattered cost columns with a single tblSiteCostLine fact table (Site × Unit Cost × Quantity) linked to tblSites and tblUnitCosts. Every cost is now Quantity × Unit Cost, fully traceable, and the catalogue can be repriced in one place. Totals were reconciled against the original model.
3. **Query layer (SQL).** Built reusable queries for cost roll-ups, schedule variance, geographic and technical distributions, and a parameterized site search.
4. **Report layer.** Designed the Site Profile report: a one-page-per-site view composed of 7 subreports (dates, regulatory, permits, infrastructure, electrical, network, cost lines).
5. **Automation (VBA).** Added Visual Basic routines to generate reports on demand and export them to PDF per site or per province, standardizing the recurring reporting cycle.
6. **Validation and reporting.** Cross-checked totals, tested edge cases (null actual dates, zero-quantity lines), and issued regular reports with findings and conclusions.

4.3 Key SQL (excerpts)

Total CAPEX per site

```
SELECT s.SiteID, s.SiteCode, s.SiteName, s.Province,
```

```
       Sum(scl.Quantity * uc.[Unit Cost USD]) AS TotalCostUSD
FROM (tblSites AS s
      INNER JOIN tblSiteCostLine AS scl ON s.SiteID = scl.SiteID)
      INNER JOIN tblUnitCosts AS uc ON scl.UnitCostID = uc.UnitCostsID
GROUP BY s.SiteID, s.SiteCode, s.SiteName, s.Province
ORDER BY Sum(scl.Quantity * uc.[Unit Cost USD]) DESC;
```

Schedule variance with status classification

```
SELECT s.SiteCode, s.SiteName, m.MilestoneName,
       d.[Finish Plan], d.[Finish Real],
       DateDiff("d", d.[Finish Plan], d.[Finish Real]) AS VarianceDays,
       IIf(d.[Finish Real] Is Null, "Not Completed",
           IIf(DateDiff("d", d.[Finish Plan], d.[Finish Real]) > 0,
               "Late", "On Time")) AS Status
FROM (tblDates AS d
      INNER JOIN tblSites AS s ON d.SiteID = s.SiteID)
      INNER JOIN tblMilestones AS m ON d.Milestone = m.MilestoneID
ORDER BY DateDiff("d", d.[Finish Plan], d.[Finish Real]) DESC;
```

Parameterized site search (blank parameter = all records)

```
SELECT s.SiteID, s.SiteCode, s.SiteName, s.Province, s.Municipality,
       at.AccessTypeName, o.OwnerCode
FROM (tblSites AS s
      INNER JOIN tblAccessType AS at ON s.AccessTypeID = at.AccessTypeID)
      INNER JOIN tblOwner AS o ON s.OwnerID = o.OwnerID
WHERE (s.Province = [Enter Province or leave blank:]
       OR [Enter Province or leave blank:] Is Null)
  AND (at.AccessTypeName = [Enter Access Type or leave blank:]
       OR [Enter Access Type or leave blank:] Is Null);
```

4.4 Key VBA (representative automation module)

```
' Export the Site Profile report to PDF for every site in a province.
Public Sub ExportProvinceReports(ByVal strProvince As String, _
                                 ByVal strFolder As String)
    Dim rs As DAO.Recordset
    On Error GoTo ErrHandler

    Set rs = CurrentDb.OpenRecordset( _
        "SELECT SiteID, SiteCode FROM tblSites WHERE Province = '" & _
        Replace(strProvince, "'", "''") & "' ORDER BY SiteCode")

    Do While Not rs.EOF
        DoCmd.OpenReport "rptSiteProfile", acViewPreview, , _
                         "SiteID = " & rs!SiteID, acHidden
        DoCmd.OutputTo acOutputReport, "rptSiteProfile", acFormatPDF, _
                       strFolder & "\SiteProfile_" & rs!SiteCode & ".pdf"
        DoCmd.Close acReport, "rptSiteProfile", acSaveNo
        rs.MoveNext
    Loop

    rs.Close
    MsgBox "Reports exported to " & strFolder, vbInformation
    Exit Sub
ErrHandler:
    MsgBox "Export failed: " & Err.Description, vbExclamation
End Sub
```

4.5 Tools

Microsoft Access (ACE engine) · SQL (Access/Jet dialect) · Visual Basic for Applications · Entity-relationship modelling · PMBOK cost and schedule baselines · Excel (data staging and dashboard front-ends)

5. Results / Impact

All figures below are computed directly from the database.

Delivered system

- One source of truth for 100 sites, 4,600 cost lines, and 1,100 schedule records, with 16 defined relationships and no orphaned site or schedule records.
- Fully traceable costing: every dollar of the US$11.48 M CAPEX baseline resolves to quantity × catalogue price, replacing about 27 hard-coded cost columns with one fact table.
- Instant roll-ups by site, area, province, owner, and access type through saved queries.

Insights produced

Finding	Evidence
RF and Infrastructure drive 69% of CAPEX	RF US$4.49 M (39.1%) + Infrastructure US$3.40 M (29.6%) of US$11.48 M
Average site CAPEX of US$114.8 K	Range US$49.4 K to US$174.1 K; the 3.5× spread shows where cost control matters most
Cost per inhabitant covered ≈ US$116	US$11.48 M ÷ 99,320 inhabitants
Standardized structure strategy	83 of 100 sites use monopoles; mean structure height 21.1 m
Access type affects cost	Pedestrian-access sites average US$102 K vs. US$120 K for aerial-access sites
Land-tenure mix	35 sites on company-owned property, 24 public schools, 21 private, 20 local-government facilities; 44 agreements, 35 own sites, 21 leases
Schedule tightly controlled but consistently slightly late	Of 736 milestones with a recorded finish, 612 (83%) closed after plan by an average of 2.6 days; none by more than 5 days
Transmission, Infrastructure, and Permits slip the most	Average slip of 3.5, 3.3, and 3.2 days respectively
Progress snapshot	60 sites with construction finished and 45 sites ready, across 1,100 tracked milestones

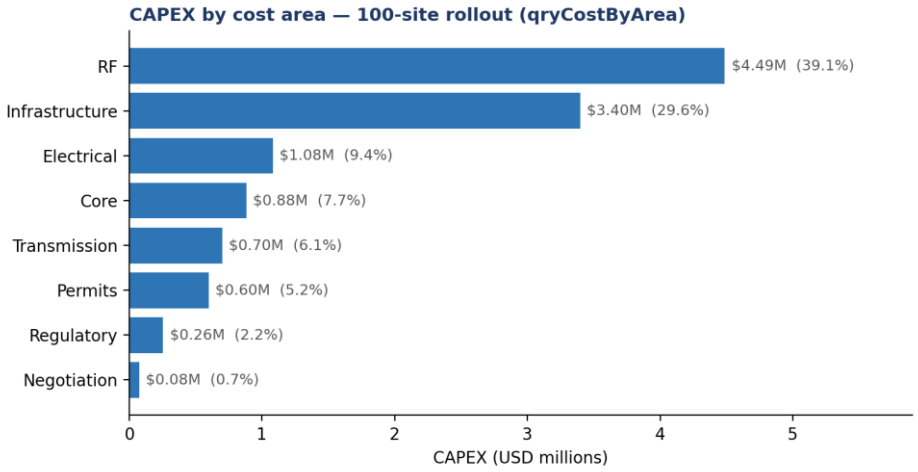


Figure 1. CAPEX by cost area, computed with qryCostByArea.

Business value

- **Faster, repeatable reporting** with PDF generation per site or province, replacing manual assembly.
- **Early-warning visibility** on milestones that slip, supporting corrective action before delays compound.
- **Better budget decisions** through transparent cost drivers and a single editable price catalogue (EUR and USD).
- **Audit-ready numbers:** one definition of cost and status, reused across every report.

6. Visuals / Artifacts

The charts below were generated from the database. Screenshots of the working Access objects should be added in the positions indicated.

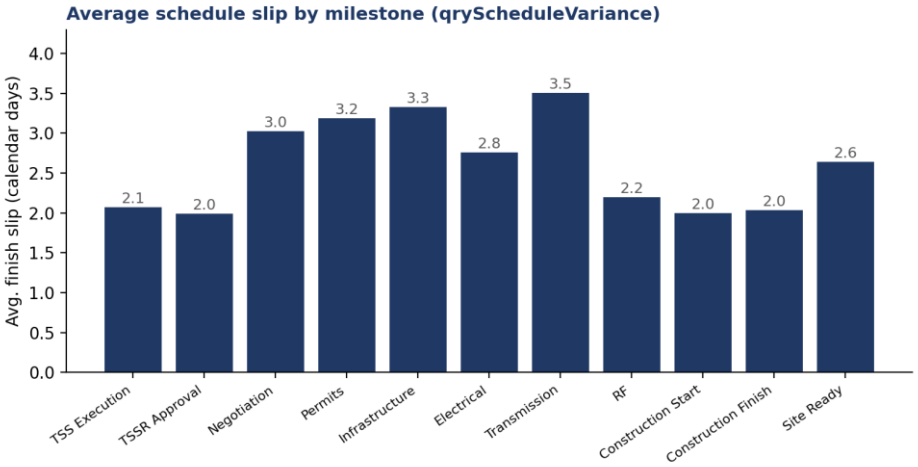


Figure 2. Average finish slip by milestone, from qryScheduleVariance.

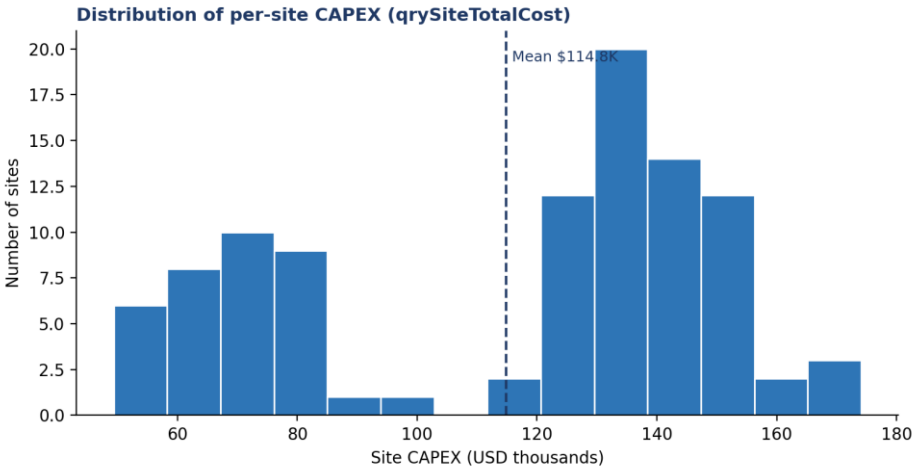


Figure 3. Distribution of per-site CAPEX, from qrySiteTotalCost.

Recommended screenshots to add

#	Visual	Why it helps
1	Relationships window / ERD	Shows data-modelling skill at a glance
2	Query design view of qrySiteTotalCost and its datasheet result	Proves SQL joins and aggregation
3	qryScheduleVariance datasheet, sorted by variance	Shows analytical thinking, not just structure
4	Site Profile report (page 1) with its subreports	The most visual deliverable; use a clean, anonymized site
5	VBA editor with the export routine	Evidence of Visual Basic skills
6	Exported PDF opened beside the Access window	Demonstrates the end-to-end automated workflow
7	Before/after cost model (scattered columns to tblSiteCostLine)	Tells the design-improvement story

#	Visual	Why it helps
8	Navigation Pane showing tables, queries, and reports	Conveys scope and organization

7. Lessons Learned

Technical

- **Model costs as facts, not columns.** A quantity × unit-price structure makes totals auditable and repricing trivial. Designing the catalogue relationship from day one avoids a mid-project redesign.
- **Enforce referential integrity early.** Lookup tables and foreign keys prevent inconsistent entry and catch errors at load time rather than at reporting time.
- **Treat nulls deliberately.** Milestones without actual dates must be classified ("Not Completed"), never silently dropped, or schedule metrics are overstated.
- **Parameterized queries beat duplicated queries.** One search query with optional parameters replaced a family of near-identical filters.

Management

- **Start from the decisions, not the data.** Defining the questions stakeholders ask first kept the database lean and the reports useful.
- **Small, consistent delays add up.** A 2–3 day average slip looks harmless per milestone but compounds across 11 sequential milestones; tracking by milestone type shows where to intervene.

Benefits to the organization

1. **A single source of truth** shared by engineering, finance, permitting, and project control.
2. **Cost transparency:** clear visibility of what drives CAPEX (RF and Infrastructure ≈ 69%) and where per-site variation comes from.
3. **Proactive schedule control** through objective, milestone-level variance reporting.
4. **Time savings and consistency** from automated, repeatable reports.
5. **Scalability:** the same model extends to additional sites by adding rows, not by redesigning the structure.

Data and figures are derived from the Terranova project database. Costs are catalogue-priced estimates (EUR converted to USD at 1.16).